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ORIGINAL ARTICLE

Research on Moisture Detection in Transformer Oil Using an Improved Neural Network Model and Broadband Ultrasonic Technology

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Abstract
The moisture content in transformer oil is a critical indicator for maintaining power system reliability. In this research, a novel model combining broadband ultrasonic detection technology coupled with an Improved Whale Optimization (IWOA) and an Elman neural network is proposed for predicting the moisture content in transformer oil. Initially, the study analyzed 240 transformer oil samples, measuring their moisture content using a moisture content meter. A custom-designed broadband ultrasonic detection platform revealed compelling correlations between ultrasonic signatures and moisture levels across samples. Subsequently, an optimized Elman neural network model was developed to predict the moisture content in transformer oil. The input features included ultrasonic amplitude, phase, and attenuation coefficients at different frequencies, while the output represented actual moisture content. Through rigorous validation, the findings indicate that the IWOA-Elman model achieves a prediction accuracy of up to 96%. The prediction accuracy meets the national standard GB/T 7600, and the integration of broadband non-destructive ultrasonic sensing technology with the IWOA-Elman algorithm establishes a new framework for online predicting of trace moisture content in transformer oil.

Keywords Transformer oil · Moisture content · Improved whale optimization · Elman neural network · Broadband ultrasonic

1 Introduction

Transformer oil as a critical component of oil-immersed transformers, plays an essential role in insulation, arc suppression, and cooling, its safety hazards have received much attention. However, during prolonged operation, insulation oil is prone to aging, with an increase in moisture content being a major factor contributing to the degradation of its insulation strength. When the moisture content exceeds a specific threshold, it can significantly impair the transformer's performance, reducing its service life or even causing catastrophic failure, rendering it inoperable. Consequently, precise detection and monitoring of the moisture content in transformer oil is of paramount importance for ensuring the reliability and longevity of transformers.

Methods for detecting moisture content in transformer oil can be broadly classified into offline and online detection [1, 2]. While offline detection methods, particularly coulometric detection, provide high accuracy, they cannot meet real-time monitoring requirements. Researchers have explored various online assessment methods for transformer oil quality. For instance, Chen Bin et al. [3] analyzed the

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impact of moisture content on insulation oil using midinfrared spectroscopy, but the spectra were influenced by other substances, leading to inaccuracies in the received data. Jiang Jun et al. [4] proposed a moisture sensing technology based on optical fiber S-taper sensors, but these sensors are challenging to manufacture and exhibit strong environmental dependence. Song Yun-Xiang et al. [5] proposed using sensors based on cross-capacitance principles, however, this approach is limited by its high susceptibility to electromagnetic interference, which affects measurement stability and accuracy. Ultrasonics first emerged in the early 20th century. Researchers led by Baêsso and colleagues employed ultrasonic parameters as a method for evaluating biodiesel quality [6], demonstrating advantages such as high accuracy and low pollution. Chen Jun-Yu et al. [7] utilized ultrasonic technology for concentration measurements in solid-liquid suspension systems, offering a new approach for monitoring slurry mixing processes. Hu Run-Run et al. [8] proposed using ultrasonics to test the high-frequency shear viscosity of peanut oil. Liu Tang-Yu et al. [9] conducted a method of ultrasonic treatment for oil sludge, addressing the impact of solid particles and oil characteristics on separation efficiency. Dengaev et al. [10] investigated innovations in oil through ultrasonic-assisted chemical transformation of oil components. Wen Hai-Ting et al. [11] provided a systematic review of ultrasonic technology enhancement in biological wastewater treatment. However, when ultrasonic technology is directly applied to transformer oil, the acoustic feature stability significantly decreases due to oil viscosity characteristics and impurity interference. The core challenge in current research lies in the highly nonlinear and time-varying relationship between multimodal acoustic parameters and trace moisture content, making it difficult for traditional physical modeling methods to construct universally applicable analytical models.

Based on the above research background and challenges, isolated non-destructive ultrasonic detection research is far from sufficient. The integration of neural networks with ultrasonic detection offers a promising solution by leveraging artificial intelligence's powerful pattern recognition capabilities to enhance detection accuracy and reliability [12–15]. More importantly, the uncertainty modeling methods proposed by Abbasi et al. [16–19] in measurement systems, such as the point estimation method, fuzzy logic method, scenario generation, and multi-objective framework, provide critical theoretical support for improving the reliability and accuracy of transformer oil moisture content measurement. Therefore, this study utilizes broadband ultrasonic technology in conjunction with an Improved Whale Optimization algorithm-Elman neural network (IWOA-Elman) to detect moisture content in transformer oil. On this foundation, the IWOA algorithm enhances the traditional whale optimization algorithm by incorporating adaptive weights and local search strategies, effectively optimizing the Elman neural network's parameters and improving its convergence speed and accuracy. The collected ultrasonic echo signals undergo Fourier analysis, considering the amplitude, phase, and attenuation coefficient of the ultrasonic echo are used as inputs to the model, and the output is the actual water content. Experimental results demonstrate that this method achieves higher detection accuracy than the traditional model.

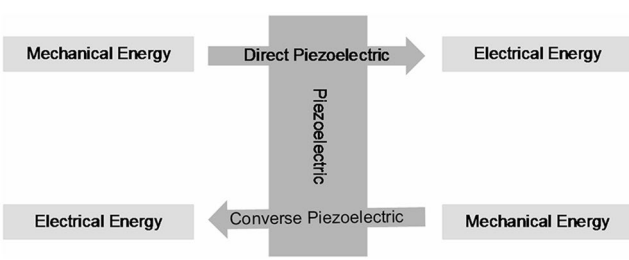

Fig. 1 Ultrasonic transducer energy conversion schematic

2 Broadband Ultrasonic Detection System for Moisture in Transformer Oil

2.1 Piezoelectric Ultrasonic Transducer

Piezoelectric ultrasonic transducers have emerged as a widely adopted technology among ultrasonic transducer types. These transducers operate through the piezoelectric effect, where specific materials generate electrical charge when subjected to mechanical stress, and vice versa [20]. This bidirectional property enables the materials to function as both transmitters and receivers of ultrasonic waves, with the energy conversion process detailed in Fig. 1. Compared to traditional large ultrasonic transducers, micromachined ultrasonic transducers offer smaller size and mass, along with reduced power consumption. To enhance the transducer's sensitivity to moisture detection in transformer oil, a DYW-E piezoelectric hydroacoustic transducer was implemented, as shown in Fig. 2. The integration of transmitting and receiving functions into a single transducer enables more effective capture of both scattered and reflected ultrasonic signals within the measured medium.

In this study, the technology's working scope includes:

I. Temperature range: Usually 25 °C ± 0.1°C;
II. Detection sensitivity: Probe drive voltage 24 V; Distance 0.3 m; Echo amplitude 500mV;
III. Frequency range: From 100 kHz to 800 kHz;
IV. Sample condition requirements: Oil samples must maintain a homogeneous state, avoiding bubbles and impurity interference;

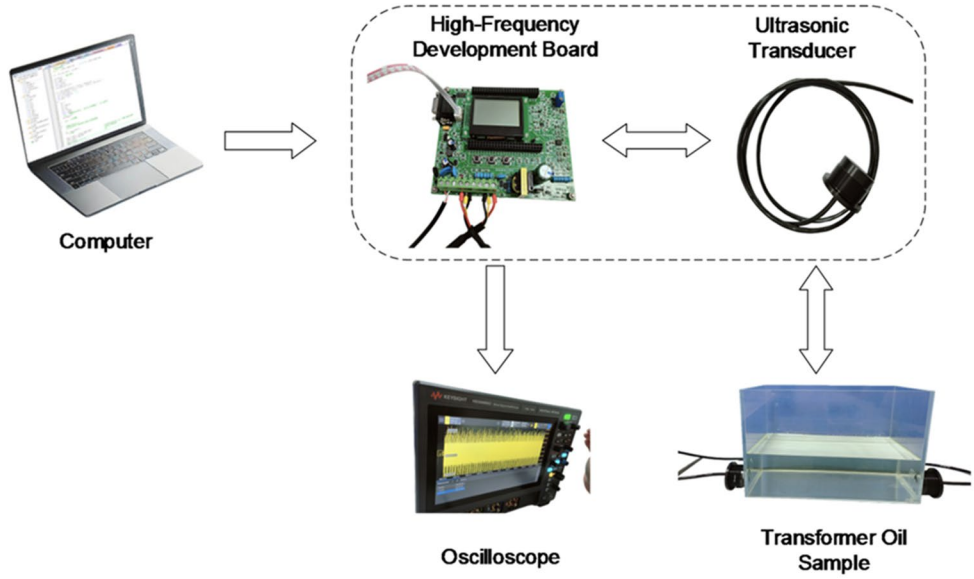

Fig. 2 Broadband Ultrasonic Detection System

Furthermore, the ultrasonic transducer should incorporate a calibration protocol: frequency verification, moisture content calibration, temperature compensation, and on-site calibration.

2.2 Broadband Ultrasonic Detection Device

The structure of the broadband ultrasonic detection system is shown in Fig. 2, it employs a quartz glass container for holding transformer oil samples, chosen for its excellent high-temperature resistance, corrosion resistance, and minimal interference with ultrasonic wave transmission. Ultrasonic transmitters and receivers are placed at both ends of the container holding the transformer oil, with transmitter and receiver 2 on the left and receiver 1 on the right. During the experimental process, ultrasonic waves are transmitted across a frequency range of 100 kHz to 800 kHz, with measurements taken at 50 kHz intervals. Each frequency measurement is repeated three times, with the median value recorded as the final reading. To ensure data accuracy, high-performance amplifiers are incorporated into the circuit design to minimize external interference. The final received data is displayed on an oscilloscope for visual analysis of the amplitude, phase, attenuation coefficient, and acoustic velocity of the received frequency.

During the operation of the broadband ultrasonic detection system, the ultrasonic wave propagation follows the principles of transmissibility and reflectivity. The emitted signal undergoes multiple interactions: a portion penetrates through the oil and reaches ultrasonic receiver 1, denoted as S1, and another part, after passing through ultrasonic receiver 1, reflects to ultrasonic receiver 2, denoted as S2. The initial ultrasonic signal introduced into the transformer oil is a sinusoidal wave, which can be recorded as Eq. (1):

$$x(t) = A\sin(2\pi f t + \phi) \tag{1}$$

Where A is the amplitude of the ultrasonic wave, f is the frequency, t is time, and ϕ is the phase. The Fourier transform is applied to the 15 frequencies of S1 and S2 to extract the maximum amplitude response and its corresponding phase response. The continuous Fourier transform signal is represented as shown in Eq. (2):

$$x(t) = \frac{a_0}{T} + \frac{2}{T}\sum_{j=1}^{\infty}\left(a_j\cos(2\pi f_j t) + b_j\sin(2\pi f_j t)\right) \tag{2}$$

Where a_0 represents the average value of the signal, T is the period of the signal, a_j is the amplitude of the cosine wave corresponding to the f_j, and b_j is the amplitude of the sine wave corresponding to the f_j.

In addition to basic wave propagation, ultrasonic waves undergo a series of attenuation responses when traveling through oil media. The attenuation process is characterized by three primary mechanisms: scattering attenuation, viscous attenuation, and thermal conduction attenuation [21–23]. They work together to influence the overall ultrasonic wave behavior.

Scattering attenuation occurs due to the scattering of sound waves caused by inhomogeneities within the medium, such as suspended particles and bubbles, which redirect energy in different directions. According to Rayleigh scattering theory, when the size of the scatterers is much smaller than the wavelength, the scattering coefficient can be expressed as in (3):

$$\alpha_s = \frac{2}{9}\pi\kappa^2 r^6 n_0 \tag{3}$$

Where k represents the ultrasonic wave number, $k=2\pi/\lambda$. Where $\lambda=c/f$, leading to $k=2\pi f/c$, and n_0 is the number of scattering particles per unit volume, $n_0=3/2\pi r^3$. Substituting these into Eq. (4):

$$\alpha_s = \frac{16}{3}\frac{\pi^4 r^3 f^4}{c^4} \tag{4}$$

Where f is frequency, and c is the propagation speed of the ultrasonic wave.

Viscous attenuation occurs when the viscous forces within the medium dissipate acoustic energy into thermal energy, causing the amplitude of the sound wave to decrease with propagation distance. Viscous attenuation can be derived using the momentum equation of the sound wave:

$$\frac{\partial V}{\partial t} = -\frac{1}{\rho}\nabla P + \nu\nabla^2 V \tag{5}$$

Where V is particle velocity, ρ is the density of the medium, P is pressure, and v is kinematic viscosity, which describes the internal viscosity of the liquid. By solving the above equation, the viscous attenuation coefficient can be expressed as in (6):

$$\alpha_\nu = \frac{2\pi^2 \nu f^2}{c^3} \tag{6}$$

Where v is the kinematic viscosity of the liquid, with units of m^2/s, f is the frequency, and c is the sound speed.

Thermal conduction attenuation occurs due to temperature fluctuations induced by sound wave propagation in the medium, where part of the acoustic energy is converted into heat through thermal conduction. By utilizing the thermodynamic equations and the energy conservation equation, it can be expressed as:

$$\alpha_h = \frac{\pi^2 \kappa f^2}{c_p \rho c^3} \tag{7}$$

Where k is thermal conductivity (unit: $W/(m{\cdot}K)$), c_P is specific heat capacity (unit: $J/(kg{\cdot}K)$), f is frequency, and c is sound speed. The total attenuation coefficient is derived from the above three attenuation coefficients:

$$a = a_s + a_v + a_h \tag{8}$$

Table 1 Transformer oil samples

Voltage level	Years in operation	Samples
10 kV	1–5	80
	5–10	40
110 kV	1–5	70
	5–10	50

For accurate measurement of attenuation coefficients in transformer oil moisture detection, a two-step system process is implemented. The first step focuses on initial calibration, utilizing standard samples with known moisture content to establish baseline measurements. The second step includes statistical analysis of repeated measurements to ensure reliability and validate the results. After the derivation process of the above formula, the main changes occurring when ultrasonic waves penetrate transformer oil are determined, and the main parameter changes of ultrasonic waves of different frequencies penetrating different transformer oil moisture content are calculated using the formula, which is used as the input data for the neural network in the following.

2.3 Detection of Moisture in Transformer Oil Based on Broadband Ultrasonic

This study examines 240 transformer oil samples collected from 110 kV transformers that have been in operation for periods ranging from 1 to 10 years. The detailed sample distribution is presented in Table 1.

According to the Chinese National Standard GB/T 7595−2017, for transformers rated at 110 kV and below, the moisture content standards are as follows: the moisture content should not exceed 20 mg/kg before commissioning and should remain below 35 mg/kg during operation. Based on the above conditions, to cover all scenarios, transformer oil samples with moisture content of 7.5 mg/kg, 15.3 mg/kg, 24.2 mg/kg, 36.2 mg/kg, and 45.0 mg/kg were selected to analyze the trend of each parameter of its moisture content at different frequencies.

Figure 3 shows the amplitude response of S1 signals received by ultrasonic receiver 1 for five oil samples with different moisture content, derived using the continuous Fourier transform formula from Sect. 2.2. Since the signals of S1 are signals that have completely passed through the transformer oil and are not affected by the ultrasonic receiver 2, it can be seen from Fig. 3 that between 100 kHz and 800 kHz, the amplitudes of the different moisture contents show an overall increase with increasing frequency. However, after 700 kHz, the amplitude suddenly decreases because high-frequency ultrasonic waves can cause waveform distortion or nonlinear energy dissipation as the frequency increases. Additionally, it is observed that at the same frequency, the amplitude gradually decreases as the water content increases.

Figure 4 shows the signal amplitude of ultrasonic waves reflected to ultrasonic receiver 2 after passing through ultrasonic receiver 1. It also exhibits a general trend where the amplitude increases with frequency for different moisture contents. However, as a reflected wave, Fig. 4 reveals that

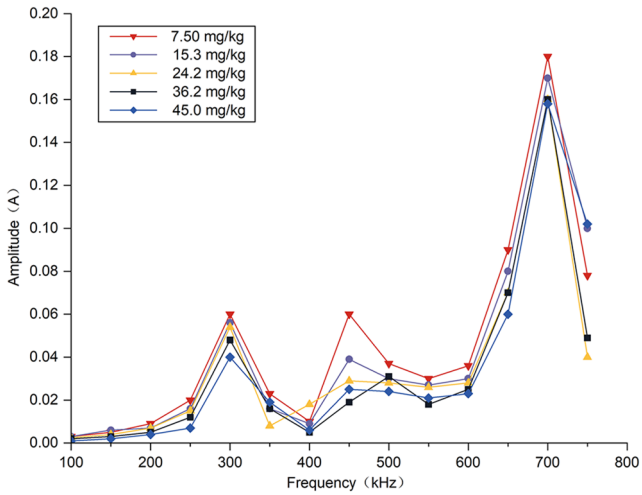

Fig. 3 S1 Amplitude Response

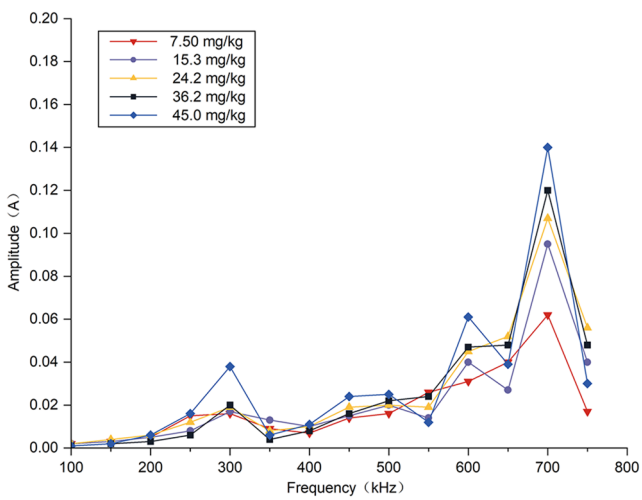

Fig. 4 S2 Amplitude Response

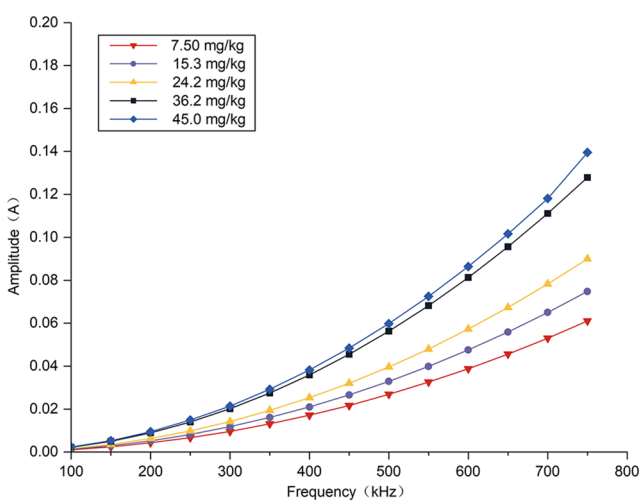

Fig. 5 Attenuation Coefficient of Broadband Ultrasonic Echo Signals

the signal amplitude of ultrasonic receiver 2 continuously increases with higher water content but sharply decreases after 700 kHz. This further indicates the close correlation between broadband ultrasonic waves and the moisture content in transformer oil.

According to the attenuation formula in Sect. 2.2, the attenuation coefficients were calculated as shown in Fig. 5. The figure indicates that as the frequency increases, the attenuation coefficient rises more rapidly for transformer oil with higher water content.

Based on the above analysis, there is a significant relationship between the moisture content in transformer oil and broadband ultrasonic signals. The variation patterns of S1 and S2 are particularly prominent in the frequency range of 200 kHz to 700 kHz. Therefore, 6 amplitudes and phases corresponding to S1 and S2 within the range of 200 kHz to 700 kHz, along with six attenuation coefficients corresponding to the frequencies, were selected as inputs for the IWOA-Elman neural network in the subsequent analysis.

3 Algorithm Design

3.1 Improved Whale Optimization Algorithm (IWOA)

The Whale Optimization Algorithm (WOA), proposed by Mirjalili et al. in 2016, is a heuristic optimization algorithm inspired by the hunting behavior of humpback whales. These whales primarily use a method called "bubble-net feeding" to catch prey [24]. The traditional WOA, despite its innovative design, exhibits several notable limitations. The algorithm's exploration mechanism relies heavily on random position updates, which can lead to premature convergence and suboptimal solutions. Therefore, its search strategy often becomes trapped in local optima, failing to effectively explore the entire solution space. To address these challenges, IWOA was developed, which accelerated the convergence speed of the network training process by incorporating adaptive parameters and modified search strategies to overcome the limitations of conventional methods. It enables the Elman network to explore a more extensive search space, starting from multiple initial points, thereby finding a more optimal solution and avoiding local optima.

The input data is denoted as X.

$$X = [A_1, \ldots, A_n, \phi_1, \ldots, \phi_n, a] \tag{9}$$

Where A_1 to A_n represent the amplitudes after the Fourier transform, ϕ_1 to ϕ_n represent the phase values, and a represents the attenuation coefficient.

First, the implementation of opposition-based learning for updating low-fitness whale positions significantly enhances the algorithm's capability to process Fourier transform data from ultrasonic echo signals. This improvement enables a more comprehensive exploration of the input data space, avoiding getting trapped in local minima. The position of the opposition individual is calculated using the formula in Eq. (10).

$$X_i^{opposite} = lb + ub - X_i \tag{10}$$

Where X_i is the position of the current individual, X_i(opposite) is the position of the corresponding opposing individual, *ub* and *lb* are the maximum and minimum values of the individual.

Second, in conventional WOA, the control parameter decreases linearly to a minimum value of 0. However, ultrasonic echo signals typically contain a variety of noises and interferences. When the parameter a is too small, the algorithm may overfit these local features while ignoring more representative global feature patterns. The amplitude and phase data often show a nonlinear relationship, and transformer oils with different moisture contents may display similar characteristics at certain frequency points. Taking into account the participation of uncertainties, Eq. (11) presents an improved *a*, which ensures that variation is smoother and more rational, and helps to balance the global search and optimization.

$$a = -(\frac{2 * t}{MaxIter})^{\gamma} + 2 \tag{11}$$

Where *t* is the current iteration count, *MaxIter* is the maximum number of iterations, γ represents the decreasing parameter.

Third, three enhanced behavioral mechanisms are introduced into the whale optimization algorithm. Particularly well suited for dealing with periodic uncertainties in signals. As shown in Fig. 6, the behavior is determined based on the random number *P* and coefficient *A*, with 0.5 as the threshold.

I. When $P<0.5$ and $A<1$, the algorithm employs proven encircling prey behavior, as shown in Eqs. (12) and (13).

$$D_{amp/phase/atten} = |C \cdot X_{best}^{*} - X(t)| \tag{12}$$

Position update equation:

$$X(t+1) = X_{best}^{*} - A \cdot D_{amp/phase/atten} \tag{13}$$

Where $D_{amp/phase/atten}$ is the distances in the dimensions of amplitude, phase, and attenuation coefficient, respectively, *C* is a coefficient parameter. $C=2r_2$, r_2 is a random number in the range 0–1, X_{best} is the optimal amplitude, phase, and attenuation coefficient at the current moment, $X(t+1)$ is the updated amplitude, phase, and attenuation coefficient for the next moment.

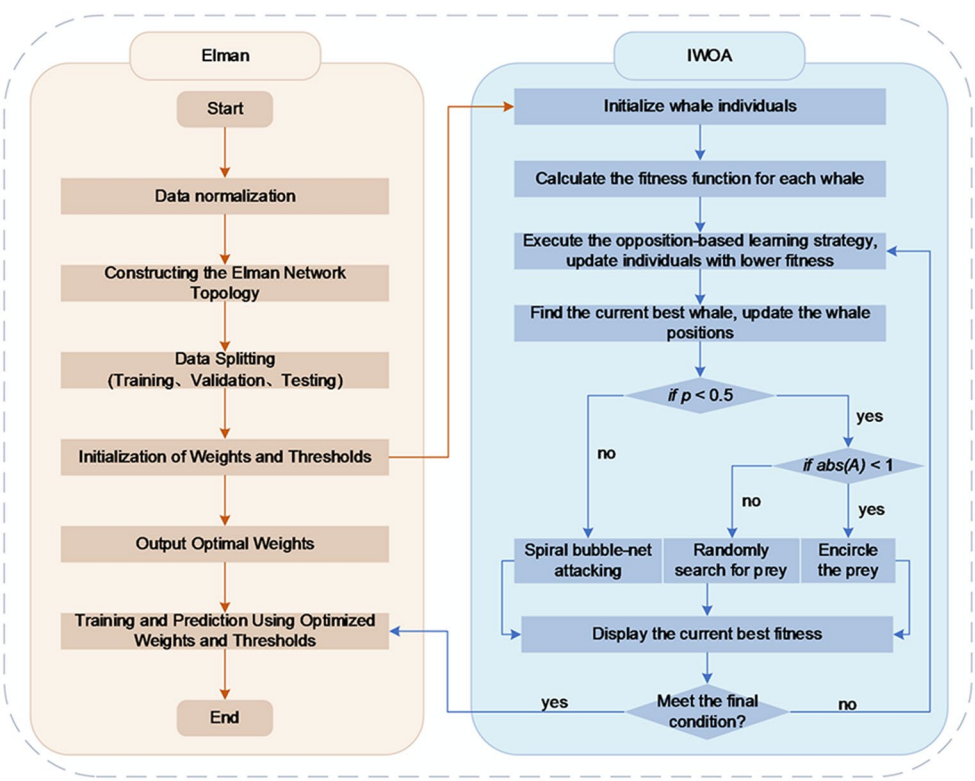

Fig. 6 The IWOA-Elman flowchart, where P and A control the probability of random behaviors

II. When $A>1$, it switches to improved random search behavior as shown in Eqs. (14) and (15).

$$D_{amp/phase/atten} = |C \cdot X_{rand} - X(t)| \tag{14}$$

Position update equation:

$$X(t+1) = X_{rand} - A \cdot D_{amp/phase/atten} \tag{15}$$

Where X_{rand} is the amplitude, phase, and attenuation coefficient of a randomly selected individual at the current moment.

III. When $P>0.5$, taking into account the periodicity and continuity of the signal characteristics, the spiral update equation can be improved as (16) and (17).

$$D'_{amp/phase/atten} = |X^*_{best} - X(t)| \tag{16}$$

Position update equation:

$$X(t+1) = D'_{amp/phase/atten} \cdot e^{bl} \cdot \cos(2\pi l) + X^*_{best} \tag{17}$$

Where b is a constant parameter for the spiral shape, l is a random number in the range [-1,1], controlling the spiral position.

Since the improved Whale Optimization Algorithm (IWOA) optimizes the weights and biases of the network, it is necessary to configure the relevant parameters first. The parameters of the Whale Optimization Algorithm, including the number of whale individuals, the maximum number of iterations, and the search bounds, are set based on the temporal nature and scale of the input data. Given that the input sample size is 30 and the input features are moderately complex, the network is considered small-scale. Therefore, the number of whale individuals is typically set between 20 and 50. After testing the stability of the network with different numbers, it is set to 30, and the number of iterations is 100.

In summary, the optimal fitness is obtained by using the improved encircling prey behavior and bubble-net attack behavior, and the optimal weights and thresholds are then used to train the subsequent network. The specific algorithm flow is shown in Fig. 6.

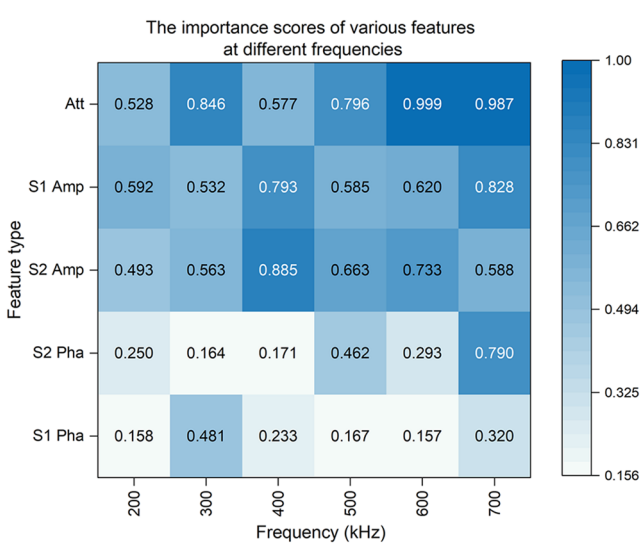

Fig. 7 The importance scores of various features at different frequencies

3.2 IWOA-Elman Neural Network Prediction Model

The Elman Neural Network is a variant of Recurrent Neural Networks (RNN), proposed by Jeffrey Elman in 1990 [25]. It adds a context layer to the standard feedforward neural network, which can store the hidden layer state from previous time steps, giving it memory capabilities. It is also better adaptable to the dynamic changes of input data, realizing the mapping of dynamic systems.

In the design of the algorithm, the data is first normalized to ensure calculations are performed on the same scale, improving the training effect and stability of the model. The normalization formula is given in Eq. (18).

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \tag{18}$$

Where X is the original data, X_{min} and X_{max} are the maximum and minimum values in the data. The normalized data X' falls within the range [0,1].

Following normalization, a Permutation Feature Importance (PFI) analysis was conducted to quantitatively assess the contribution of each input variable to moisture content prediction. As in Fig. 7, the coefficient consistently demonstrated the highest feature importance, with particular prominence at 700 kHz. The amplitude of S1 and S2 exhibited moderate contributions with discernible frequency dependence, notably around 400–500 and 700 kHz. Conversely, the phases of S1 and S2 demonstrated minimal overall contribution. This phenomenon can be attributed to water's more pronounced influence on acoustic energy attenuation compared to sound velocity, thereby diminishing the reliability of phase measurements, particularly in the complex propagation paths of reflected S2 signals. Figure 8 confirms these findings through average feature importance analysis, where the attenuation coefficient registered the highest value at 0.789, while the S1 and S2 phases recorded the lowest values, both below 0.5. Consequently, the feature inputs selected in this study are methodologically feasible.

Next, the most important aspects of creating the Elman network are setting the delay steps, the number of hidden layers, the training function, and the maximum number of iterations. Among these, the delay steps and the number of

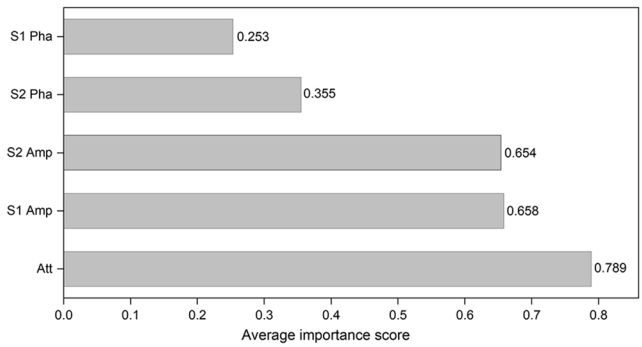

Fig. 8 Average importance score

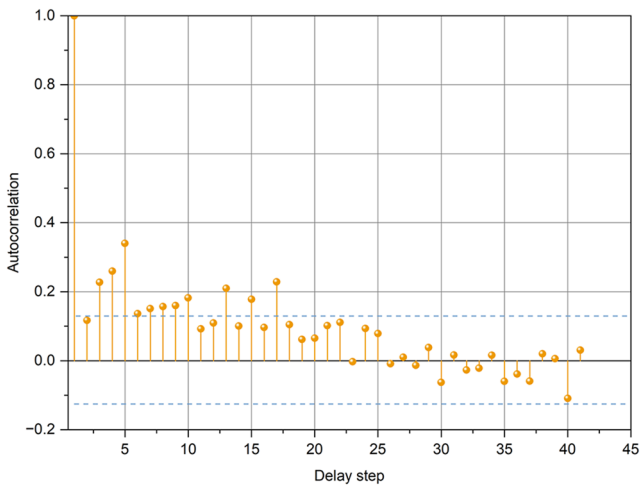

Fig. 9 Autocorrelation for different delayed steps

hidden layers are crucial during initial modeling, as selecting the appropriate values for these parameters can maximize the performance of the network. Based on the characteristics of the input data, the Autocorrelation Function (ACF) is chosen for analysis. As shown in Fig. 9, the data's autocorrelation increases between steps 5 and 10, and after step 10, the correlation gradually weakens.

The number of hidden layer neurons is roughly determined using the empirical formula (19). It is known that the input data is 30 and the output is 1. Based on this, the number of neurons is determined to be between 7 and 16 according to the empirical formula. After determining the main ranges for the two important parameters, cross-validation is used to obtain the mean squared error (MSE) corresponding to each delay step and the number of hidden neurons, as shown in Figs. 10 and 11. The MSE is minimized to 0.098 when the number of neurons is 14, and the minimum MSE of 0.165 is at a delay step of 5.

$$\sqrt{i+o}+n \tag{19}$$

Where i is the number of input neurons, o is the number of output neurons, and n is a constant between 1 and 10.

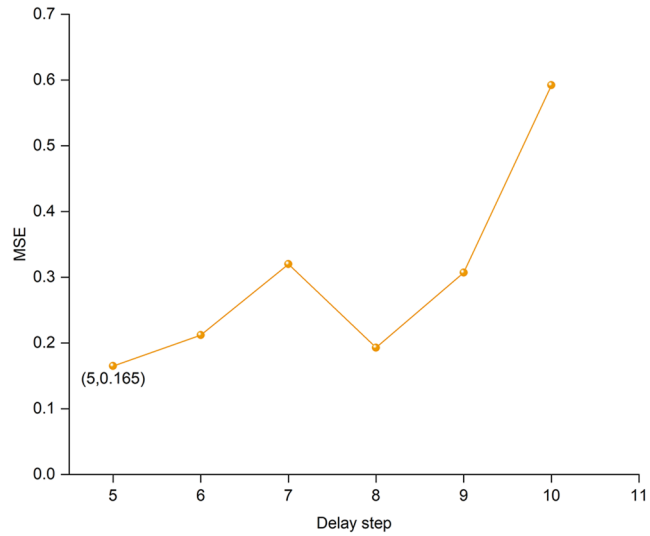

Fig. 10 MSE for delay step

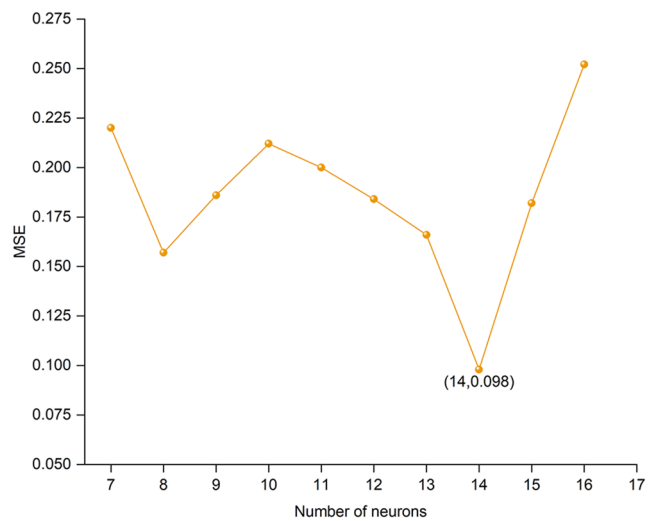

Fig. 11 MSE for different hidden neurons

Given the input data dimension of 240 × 31, which is conducive to training a small-scale network, the Levenberg-Marquardt algorithm was employed. Through iterative training and validation, the maximum number of network iterations was ultimately established at 1000. The main parameter setting for the IWOA-Elman network is presented in Table 2.

3.3 Ablation Experiment

In this study, 15 oil samples were randomly selected from a dataset of 240 transformer oil samples to evaluate the prediction accuracy of three models: Elman, WOA-Elman, and IWOA-Elman. As shown in Table 3, the selected oil samples cover the entire range of water contents in the samples. Figure 12 shows the prediction error values for the 15 samples.

Table 2 Main network parameters of IWOA-Elman

Network settings	Network parameters
Number of Whales	30
Maximum IWOA Iterations	100
Delay Steps	1:5
Number of Neurons in Hidden Layer	14
Training Function	Levenberg-Marquardt backpropagation (LM)

Table 3 Predicted values and real values for the 15 samples

Number	Actual value (mg/kg)	Elman (mg/kg)	WOA-Elman (mg/kg)	IWOA-Elman (mg/kg)
1	10.89	12.314	11.99	11.660
2	45.022	42.63	43.351	43.647
3	5.0	6.537	6.658	5.663
4	40.147	36.241	39.861	40.142
5	38.97	34.275	39.027	38.888
6	17.593	19.541	18.369	18.744
7	24.95	28.255	22.506	24.518
8	10.9	12.718	8.996	10.796
9	26.86	24.464	28.53	25.775
10	44.2	40.844	40.558	43.386
11	20.59	25.605	21.955	20.427
12	40.16	38.222	41.192	40.671
13	35.589	29.622	35.254	33.67
14	13.598	15.209	15.873	12.9
15	12.4	16.985	10.387	12.118

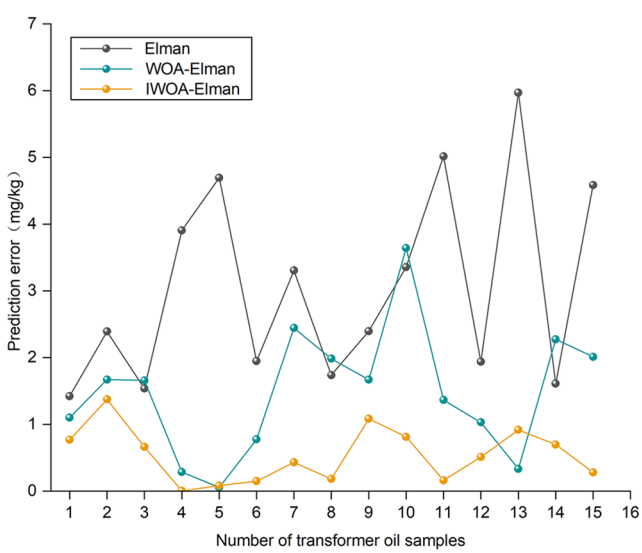

Fig. 12 Prediction errors for the 15 samples

The results reveal distinct performance characteristics for each model:

For Elman, the prediction error ranged from 2 mg/kg to 6 mg/kg, with an average prediction error of 3.054 mg/kg. Due to its simple recursive structure, the traditional Elman tends to get trapped in local optima, resulting in relatively large prediction errors. The WOA-Elman model improves overall performance by incorporating the whale optimization algorithm, reducing the average error to 1.487 mg/kg. However, its spiral search mechanism and imbalanced exploration-exploitation strategy result in noticeable prediction fluctuations (0–4 mg/kg). The IWOA-Elman model significantly enhanced prediction performance by improving the optimization algorithm through the introduction of an adaptive weight mechanism and refined parameter tuning strategies. Its prediction errors consistently remained within the 0–1 mg/kg range, with an average error of only 0.542 mg/kg, demonstrating optimal prediction effectiveness.

3.4 Contrast Experiment

Based on the nonlinear relationship between ultrasonic and moisture content, additional contrast experiments were conducted to benchmark the proposed approaches against three widely used machine learning models: LSTM, SVR, and Random Forest.

Table 4 presents a comprehensive performance comparison of computational efficiency and resource requirements. The LSTM model required 1.68 min of execution time with 8640.64 MB memory consumption, notably demanding GPU acceleration and significantly increasing hardware requirements. SVR exhibited the highest computational burden, requiring 2.39 min of execution time and 8550.29 MB of memory usage. The Random Forest implementation necessitated 2.27 min with 8519.33 MB memory utilization.

To better evaluate the predictive performance of the model, four evaluation metrics are introduced: MAPE (Mean Absolute Percentage Error), MSE (Mean Squared Error), MAE (Mean Absolute Error), and R^2 (Coefficient of Determination). Figure 13 illustrates the comparative performance metrics across all six models (Elman, WOA-Elman, IWOA-Elman, LSTM, SVR, and Random Forest). The LSTM model shows reasonable predictive ability, but the indicators are under IWAO-Elman and require GPU acceleration, which increases hardware requirements. The SVR model exhibits high computational cost with greater error margins; Random Forest demonstrates moderate performance but insufficient precision. With the addition of optimization and improved algorithms, the three metrics significantly decreased. The figure shows that the fitting effect of IWOA-Elman reaches 0.96, indicating that IWOA-Elman provides better predictions.

Moreover, to further investigate the performance robustness of the optimal IWOA-Elman model under varying environmental temperatures encountered in practical applications, its prediction performance was evaluated across the 10 °C to 60 °C range, as shown in Figs. 14 and 15. Results indicate that while the model maintained stable and low

Table 4 The training cost of each model (Note, IW-E: IWOA-Elman, W-E: WOA-Elman, R-F: random Forest)

Performance metric	IW-E	Elman	W-E	LSTM	SVR	RF
Total Execution Time	1.44 min	1.58 min	1.49 min	1.68 min	2.39 min	2.27 min
Memory Usage	8220.29 MB	8600.56 MB	8734.77 MB	8640.64 MB	8550.29 MB	8519.33 MB
Data Loading Time	0.1773s	0.1958s	0.117s	0.98s	0.217s	0.1461s
Network Creation Time	0.1651s	0.1372s	0.189s	0.60s	1.01s	1.076s
Initial Network Training Time	0.54 min	0.83 min	0.64 min	0.5 min	0.86 min	0.557 min
CPU Usage	75%	35.5%	62%	/	71.6%	62.8%

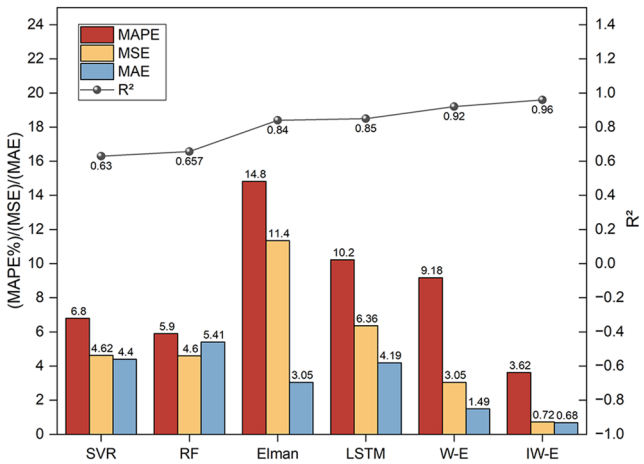

Fig. 13 Four performance metrics for each model (Note, IW-E: IWOA-Elman, W-E: WOA-Elman, R-F: Random Forest)

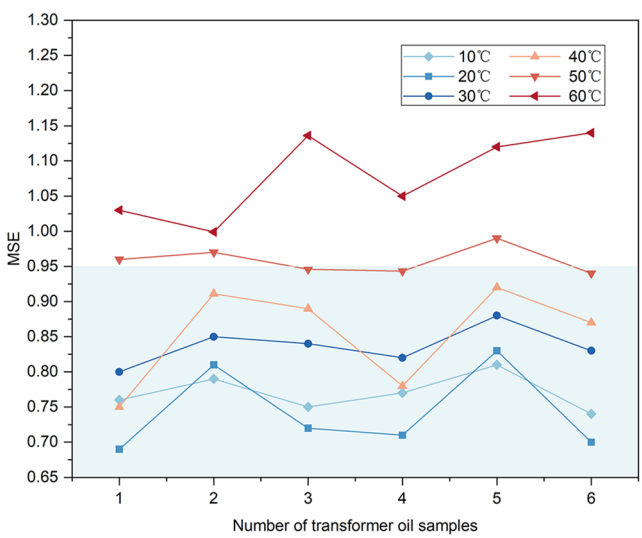

Fig. 14 MSE of the model at different temperatures

MSE and MAPE values between 10 °C and 50 °C, comparable to its high accuracy under standard conditions, prediction errors increased significantly above 50 °C (at 60 °C). This finding must be contextualized by the sensor's intended application. Since free water tends to settle at the transformer tank bottom, the sensor is designed for deployment at this key location. Based on relevant standards GB/T 1094.2–2013 and typical thermal behavior, the bottom oil

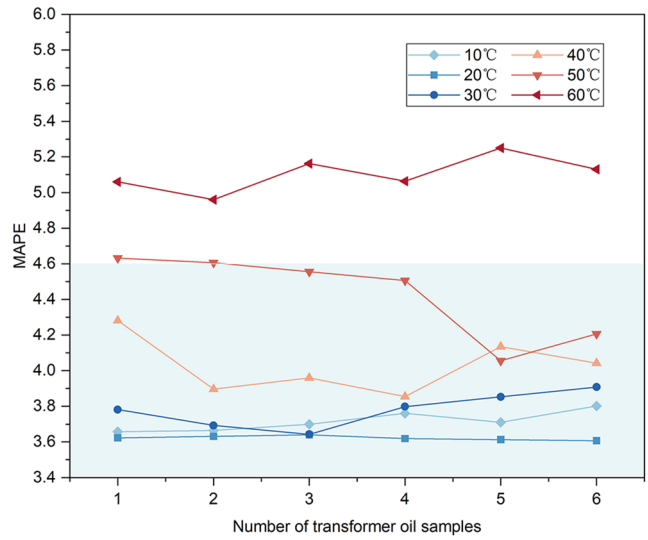

Fig. 15 MAPE of the model at different temperatures

temperature generally stays below 50 °C under most conditions and below 65 °C even at high loads, significantly cooler than the top oil. Consequently, although model performance declines above 50 °C, its high accuracy and stability within the 10–50 °C range adequately cover the predominant operating temperatures at the intended installation site. This validates the proposed model's suitability for the specific application.

4 Conclusion

This paper investigates the non-destructive detection of moisture content in transformer oil using broadband ultrasonic technology enhanced by an improved algorithmic model. Initially, traditional moisture detection methods were applied to 240 transformer oil samples under environmental conditions of 25 ± 0.1℃. Subsequently, broadband ultrasonic scanning was conducted on oil samples with differing moisture content. The Fourier transform is employed to extract characteristic information from the ultrasonic echo data, including amplitude, phase, and attenuation coefficient. This characteristic information was used to develop an optimized predictive model.

The results prove that the model achieved an accuracy of 96%, with all performance metrics reaching optimal levels. The prediction errors were generally distributed between 0 and 1 mg/kg. This study provides a reliable technical solution for condition prediction of power equipment, holding substantial importance for improving power system reliability. Further research is recommended to explore the model's adaptability under various operating conditions and investigate the possibility of extending this method to predict other transformer oil parameters.

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Author Contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Zhao-Yu Qin, Sa-Sa Kong and Guo-Zhi Zhang. The first draft of the manuscript was written by Sa-Sa Kong and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data Availability All authors are requested to make sure that all data and materials as well as software application or custom code support their published claims and comply with field standards. The data supporting this study's findings are available from the corresponding author upon reasonable request.

Declarations

Competing Interests Zhao-Yu Qin, Sa-Sa Kong, Guo-Zhi Zhang, Si-Da Xu, Lei Yang, and Jia-Bin Huang declare that we have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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